

# Muhammad Hariz Hazril

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## PROFESSIONAL SUMMARY

Recent graduate with a Bachelor of Science (Hons.) in Pure Physics and a Minor in Computer Science, offering hands-on experience through an internship in Nuclear Safeguards at the Department of Atomic Energy Malaysia. Skilled in software development, data analysis, and system design, with proficiency in Python, C++, Java, SQL, and web frameworks such as Flask, Oracle APEX, and Bootstrap. Adept at problem-solving, critical thinking, and collaborating within interdisciplinary teams on projects involving computational modelling, database management, and network security. Seeking an opportunity to leverage academic knowledge and practical project experience in software development, with a strong commitment to contributing to innovative solutions and advancing technology outcomes.

## EDUCATIONAL BACKGROUND

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|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10/2021 – 09/2025<br>Gelugor, Penang Island | <b>Bachelor of Science (Hons.) Pure Physics (Major) &amp; Computer Science (Minor).</b><br><i>Universiti Science Malaysia (USM)</i><br><b>CGPA:</b> 3.04<br><b>Thesis on</b> "Random Walk Radioactive Nuclides Transport in Underground Water".<br><b>Among subjects taken:</b> Computational Physics, Programming Methodology & Data Structures, Foundation of Software Engineering, Database Organisation & Design, Principles of Computer Networks & Information Security, Electronics I, Electronics II, Electricity and Magnetism, Solid State Physics. |
| 08/2020 – 05/2021<br>Changlun, Kedah        | <b>Physical Science Stream (1" Year Programme, Module II).</b><br><i>College Matriculation Kedah (KMK)</i><br><b>CGPA:</b> 3.83 <b>Muet:</b> Band 3                                                                                                                                                                                                                                                                                                                                                                                                          |

## WORK EXPERIENCE

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|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03/2025 – 09/2025<br>On-Site | <b>Nuclear Safeguards (Intern)-Nuclear Installation Division</b><br><i>Atom Malaysia (Department of Atomic Energy Malaysia)</i> <ul style="list-style-type: none"><li>Assisted in testing and validation of automated digital systems for nuclear facility data reporting.</li><li>Designed and implemented Python-based automated form validation to detect and resolve submission inconsistencies.</li><li>Developed a Flask-based testing interface to verify metadata accuracy and improve data traceability.</li><li>Conducted system debugging, documentation, and performance testing for in-house regulatory applications.</li></ul> |
| 06/2024 – 08/2024<br>Remote  | <b>Virtual Assistant Specialist in Data Analytics</b><br><i>Freelancing</i> <ul style="list-style-type: none"><li>Automated testing workflows for Excel-based datasets using Python and Power BI.</li><li>Conducted data validation and visualization to ensure test result integrity.</li><li>Created automated scripts to cross-check dataset consistency, reducing manual review time by 40%.</li></ul>                                                                                                                                                                                                                                   |

## PROJECT EXPERIENCE

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|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10/2023 – 07/2024 | <b>Random Walk Radioactive Nuclides Transport in Underground Water</b><br><i>Bachelor's Final Year Project</i> <ul style="list-style-type: none"><li>Developed and implemented a Random Walk Particle Tracking (RWPT) model in Python to simulate radionuclide</li><li>Applied Monte Carlo methods and numerical techniques to model advection–dispersion processes, integrating stochastic random walk theory with hydrogeological principles.</li></ul> |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

- Validated the RWPT model against the industry-standard MODFLOW 6 and MT3D-USGS solute transport simulators, achieving a high accuracy level with <1% discrepancy.
- GitHub Link: [hariz-harisun.github.io/RWPT](https://github.com/hariz-harisun/RWPT) (Password: CVHariz26)

05/2024 – 06/2024

### **Enterprise Network Design & Security (Cisco Packet Tracer)**

*Coursework Project | Principle of Computer Network & Information Security*

- Designed and simulated a secure enterprise network using Cisco Packet Tracer, implementing subnetting, structured LAN design, and configuring ~90 RJ45 ports and Wi-Fi access points.
- Configured VLANs and Access Control Lists (ACLs) to segregate internal staff traffic from visiting clients, enforcing role-based access and network permissions.

07/2025 – 09/2025

### **Additional Protocol (AP) Declaration Helper Form System**

*Internship Project | Atom Malaysia*

- Developed a Flask-based web application (Python, Flask, Bootstrap, jQuery) to efficiently search and filter research publications from the CrossRef API, with a focus on structured metadata extraction and research data management.
- Designed an interactive interface with dynamic filtering and session-based result storage, enhancing user control over literature search processes.

04/2024 – 05/2024

### **University Library Database Management System**

*Coursework Project | Database Organisation & Design*

- Led team to design and model a normalized relational schema in SQL for a Library Management System, enforcing primary/foreign key constraints and referential integrity.
- Implemented role-based access control (RBAC) through SQL-based privilege assignments for multiple user roles (students, staff, librarians).
- Built a working prototype in Oracle APEX, demonstrating interactive forms for registration, SQL-driven search pages, and automated report generation.

## **SKILLS**

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**Programming & Web:** Python (Flask, NumPy, SciPy, Pandas, Matplotlib, Scikit-learn), C++, Java, R, HTML5, CSS, JavaScript, PHP.

**Database & Cloud Tools:** SQL (MySQL, Referendum Integrity, Constraints), Oracle APEX, REST APIs, Power BI.

**Software & Simulation:** Computational Modelling, Numerical Methods, Monte Carlo, Tinker CAD, Easy68K, MT3D-USGS, MODFLOW 6, Mathematica.

**Development Tools:** Version Control (GitHub/GitLab), Anaconda, Arduino IDE, Raspberry Pi IDE, Microsoft Office Suite, Google Workspace, Microsoft Teams / Zoom.

**Core Competencies:** System Design & Architecture, Test-Driven Development (TDD), Data Analytics, Role-Based Access Control (RBAC), Network Security Principles.

**Soft Skills:** Leadership & Teamwork, Problem-Solving & Critical Thinking, Adaptability & Self-Learning, Attention to Detail.

## **AWARDS & RECOGNITIONS**

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**Dean's List Award for achieving a CGPA above 3.5 in semesters 1, 2, & 8 | USM**

*Issued by Deputy Dean (Academic, Career & International) School of Physics.*

**Best Freelancer Award (Male) | MDEC-USM 2024**

*Issued by MOHE-MDEC Global Outsourcing Talent (GOT) Bootcamp.*

**Computational Thinking Award | APS Negeri Selangor 2019**

*Issued by Sektor Sumber & Teknologi Pendidikan, Jabatan Pendidikan Negeri Selangor.*

## **LEADERSHIP EXPERIENCE**

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**Deputy President (NYDP) – Islamic Association, Universiti Sains Malaysia (USM)**

**Assistant Secretariat – Safety Analysis Report (SAR) Workshop**

**Facilitator – TECH Start: Life Skills Bootcamp**

**Event Crew – Coolest Project Malaysia 2024 (Penang Science Cluster)**

**Event Coordinator & Facilitator – USM STEM Week 2024 (School of Physics, USM)**